

More Tokens, Same Megawatts

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OCP PLATINUM Member

Johnson Controls & OCP: Partners Since 2021

From early contributor to Platinum member — collaborating across the open compute ecosystem

2021 — **Joined OCP**
Became a contributing member of the OCP community

→ **2023** — **Scaled Involvement**
Active across multiple project workstreams

→ **2025** — **Platinum Member**
Recognized among OCP's top-tier corporate members

COLLABORATING ACROSS MULTIPLE WORKSTREAMS

Data Center Facility

Contributing design input on next-gen facility and rack power/cooling interfaces.

Thermal & Liquid Cooling

Sharing chip-to-facility thermal management expertise across OCP working groups.

Hardware Management

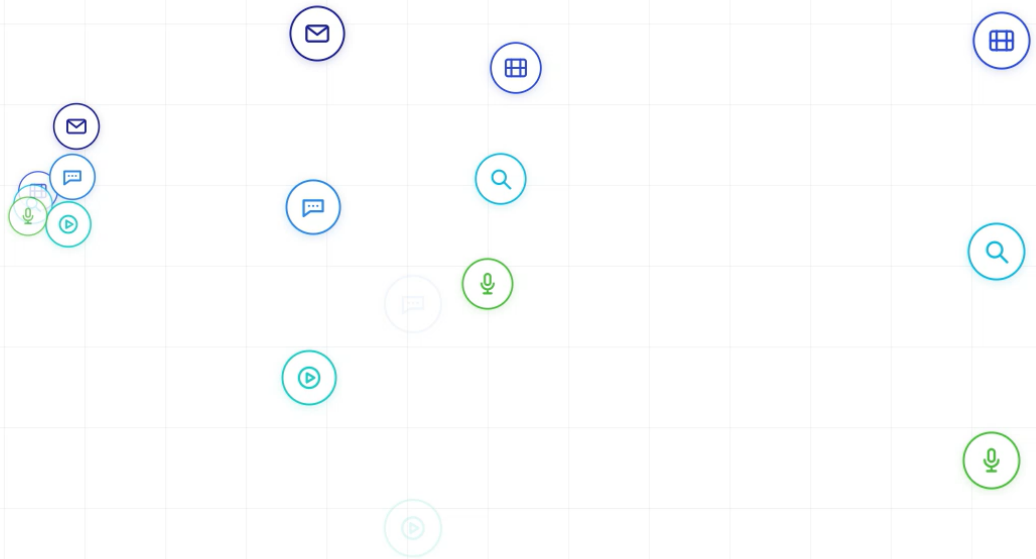
Supporting open telemetry standards between IT and facility systems.

Sustainability

Advancing efficiency benchmarks for high-density, AI-scale data centers.

One shared goal: open, efficient, sustainable infrastructure for the AI era.

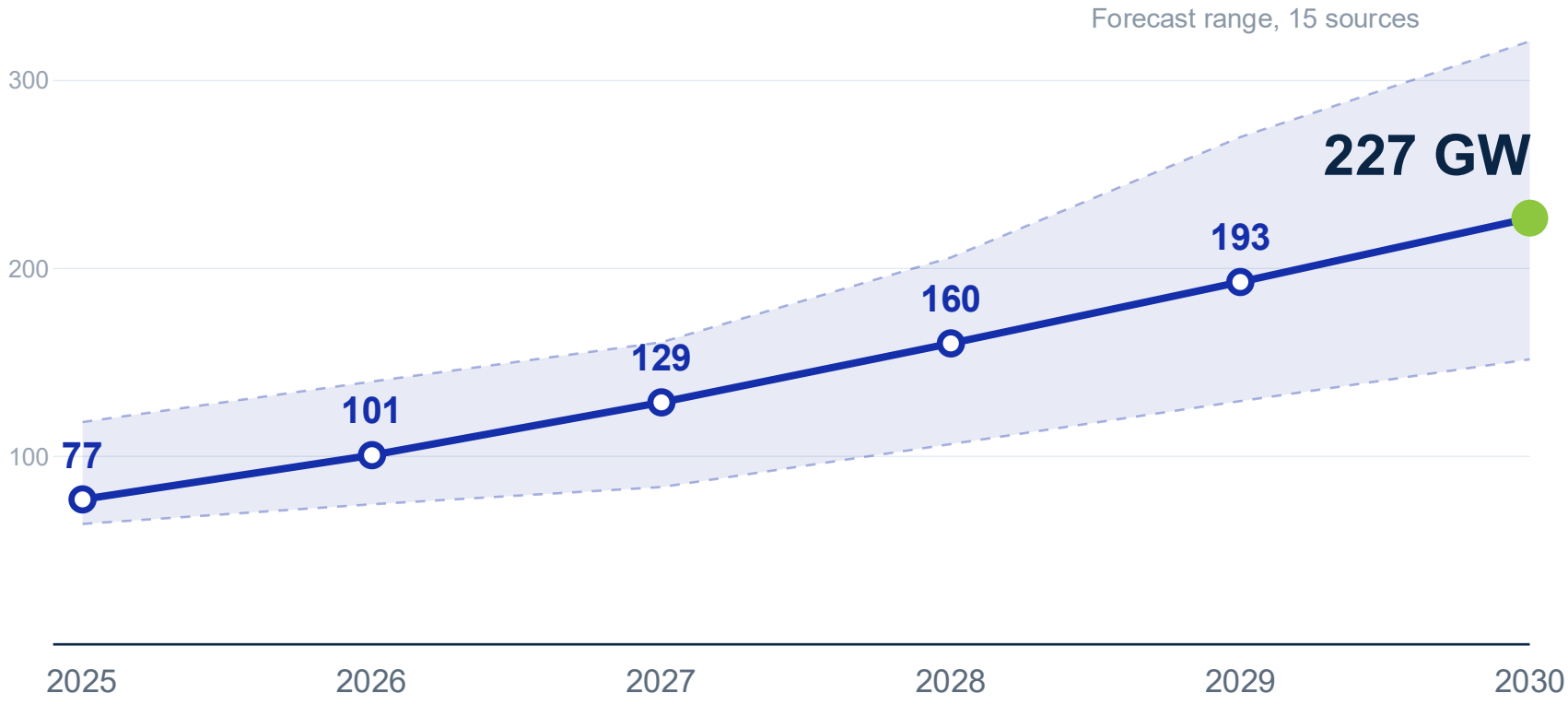
The Data Deluge Never Stops



Data center power demand triples by 2030

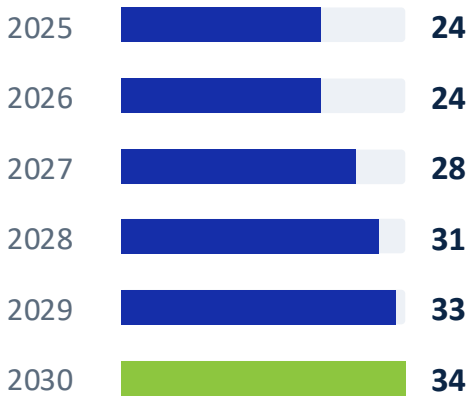
GLOBAL DATA CENTER CAPACITY (GW)

We are Going to Need a Bigger Boat!!





NEW CAPACITY ADDED EACH YEAR



GW per year, the annual build rate grows by half again over the period.

3.0× capacity, 2025 to 2030

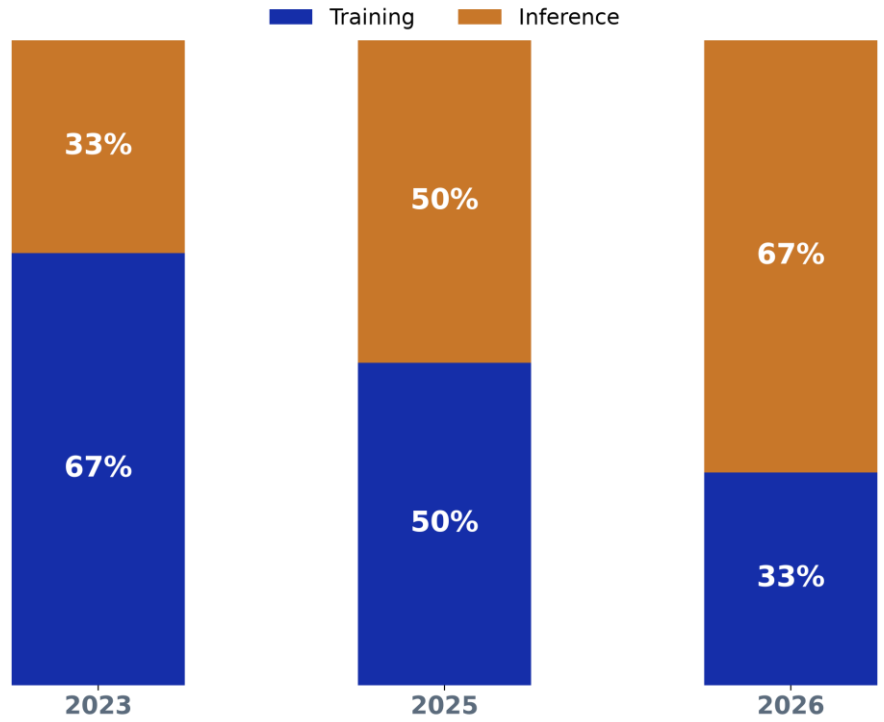
+150 GW added

24% CAGR

152–321 GW

Inference Takes Over

Share of AI compute: training vs. inference, 2023–2026



LIFETIME COMPUTE COST

80–90%

of a deployed model's total lifetime compute cost is inference training is one-time, inference runs for the product's life

2023 → 2026: inference grew from about a third of AI compute to about two-thirds.

Why: training a model happens once; inference happens every time it's used and usage keeps compounding as more products ship on top of the same models.

Sources: Deloitte, 2026 TMT Predictions; Gartner (inference share of AI-optimized IaaS spend).

14 Years: Supercomputer to Rack



IBM Blue Gene/Q

2011-2012 · 840 TFLOPS FP64 peak (4-rack)

FOOTPRINT	69 sq ft (excl. service clearance)
POWER DRAW	320 kW typical (max 400 kW)
COST	$10.4M (~$14.6M in 2025 $)
MEMORY	64 TB across 4 racks



NVIDIA GB300 NVL72

2025-2026 · up to 1,440 PFLOPS FP4 sparse (1 rack)

FOOTPRINT	6.9 sq ft
POWER DRAW	120-140 kW per rack
COST	$3-4M per rack
MEMORY	37 TB (20 TB HBM3e + 17 TB LPDDR5X)

Computation Efficiency: Space & Power

IBM Blue Gene/Q (2011-2012) → NVIDIA GB300 NVL72 (2025-2026)

COMPUTE DENSITY

~17,000× denser

12.1 TeraFLOPS/sq ft → up to 209,000 TeraFLOPS/sq ft (FP4 sparse)

ENERGY EFFICIENCY

~4,200× more efficient

2.6 GFLOPS/W → up to 11,000 GFLOPS/W (FP4 sparse)

POWER DRAW

Lower, despite far more compute

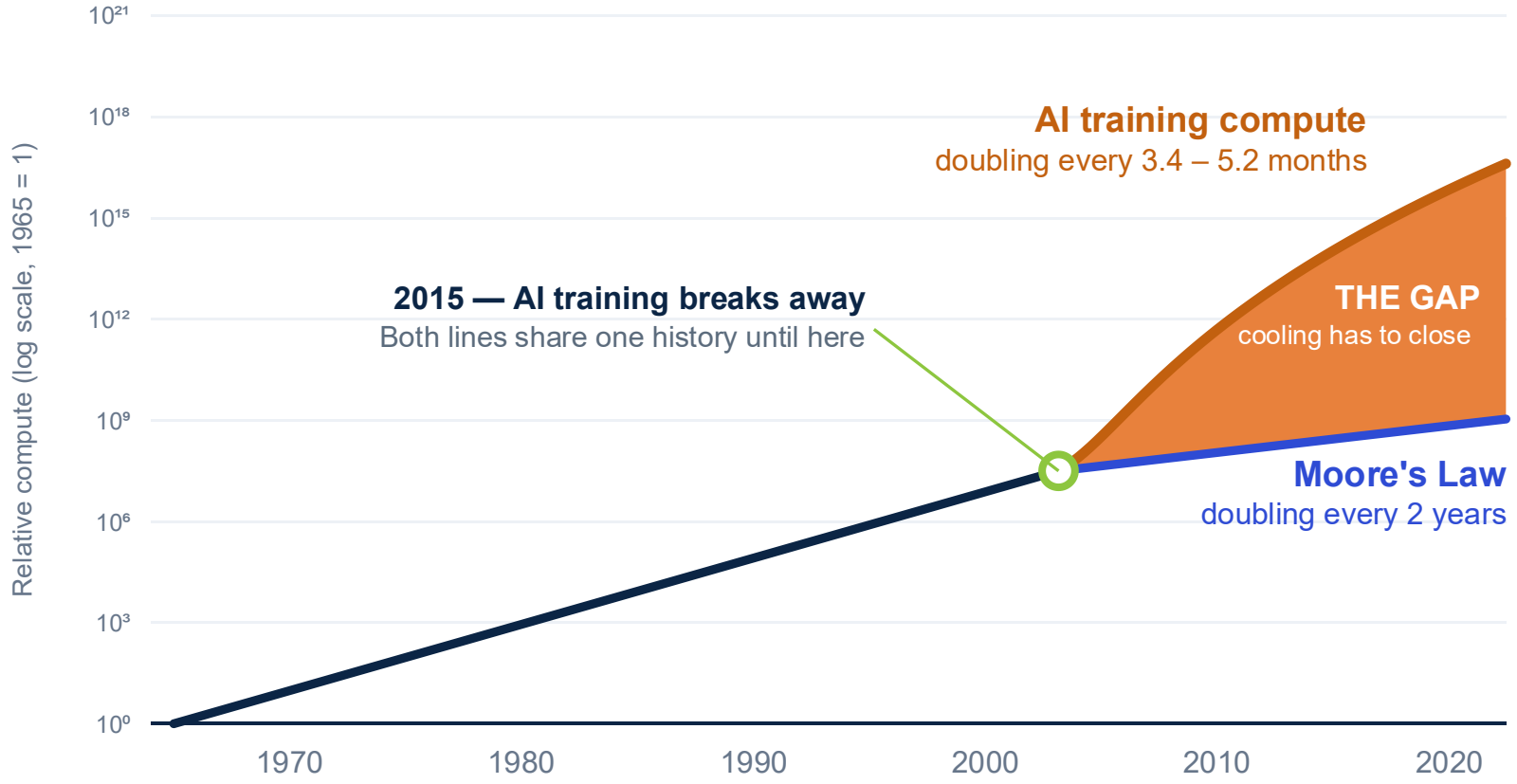
~320 kW → 120-140 kW TDP for the whole rack

14 years: the same power budget that ran one cabinet-scale supercomputer now fits inside a single rack — with room to spare

Note: BG/Q's numbers are FP64 (double precision); GB300's headline numbers are FP4/FP8/BF16 (AI-optimized, lower precision) — directionally real, not apples-to-apples on precision.

Moore's Law Isn't Keeping Up

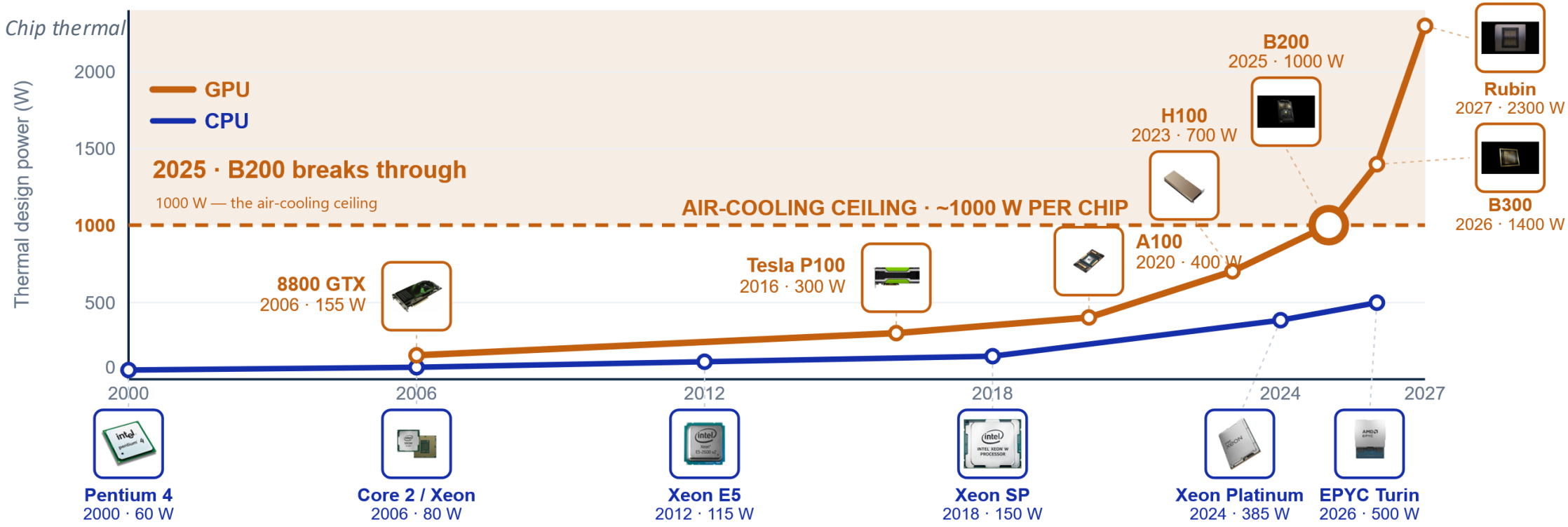
The same trend since 1965 — until AI training broke away



Sources: OpenAI, "AI and Compute" (2018); Epoch AI, compute trend data (2020–present).

Hardware alone can't keep pace — denser racks and liquid cooling had to close the gap Moore's Law couldn't

Why Air Ran Out of Runway



3,500× By volume, water carries 3,500× the heat capacity of air

+17% higher throughput (stress test)

–16% power draw (1 kW per node)

–1.4% training time (real workloads)

Once a chip passes 1000 W, liquid isn't an upgrade — it's the only way to run it

Liquid vs. air results: NVIDIA whitepaper, Oct 2025 (8 × H100 systems)

Thermal Management: The Full Cooling Taxonomy

Multiple JCI data center designs and products can be leveraged to manage heat produced by IT infrastructure

Legend

- Heat rejection
- Liquid-cooling / hybrid
- Air-cooling

IT infra.
Generates heat

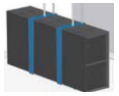


Air cooling

Aisle Containment
System separating hot and cold airflow



In Row / Rack Cooling
Cooling units placed between server racks



Liquid cooling

Direct-to-Chip
Cold plates attached directly to CPUs / GPUs



Rear Door Heat Exchanger (RDHx)
Fans and Coil mounted to back of a rack



Immersion
Servers fully submerged in dielectric liquid



Heat capture

Heat transfer

Computer Room Air Handler (CRAH)
Circulates cooled air within the server room



Cooling Distribution Unit (CDU)
Circulates coolant to servers and racks



Heat rejection

Air Cooled Chiller
Rejects heat into air using coils and fans



Water Cooled Chiller
Rejects heat to water and a cooling tower



Cooling Tower
Rejects heat to environment via evaporation



Dry Cooler
Rejects heat without evaporating water



Why asking AI a question uses electricity

1 You ask a question

You type something to the AI in ordinary words.

2 It gets cut into tokens

How | is | the | weath | er | today | ?

Each small piece is called a **token** a few letters, not always a whole word.

3 The LLM thinks

The AI model — an **LLM** — reads the tokens and works out the next one, one at a time.

4 Big computers do the work

Every piece runs on powerful machines in a data centre. They use electricity and get hot.

5 The heat has to be removed

Cooling takes the heat away — and the cooling uses electricity too.

6 Your answer appears

Token by token. A longer answer means more tokens — and more power.

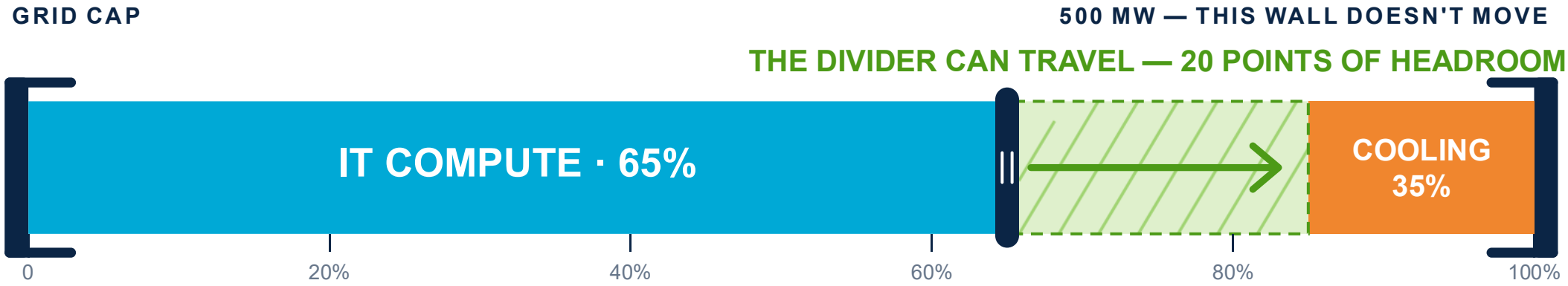
More tokens = more power consumption

Every extra token is one more round of thinking, power and cooling

A normal answer is about 500 tokens — the circle runs 500 times

So the number worth improving is tokens per joule — how much AI you get for each unit of energy

The Power Budget Is Fixed. The Split Isn't



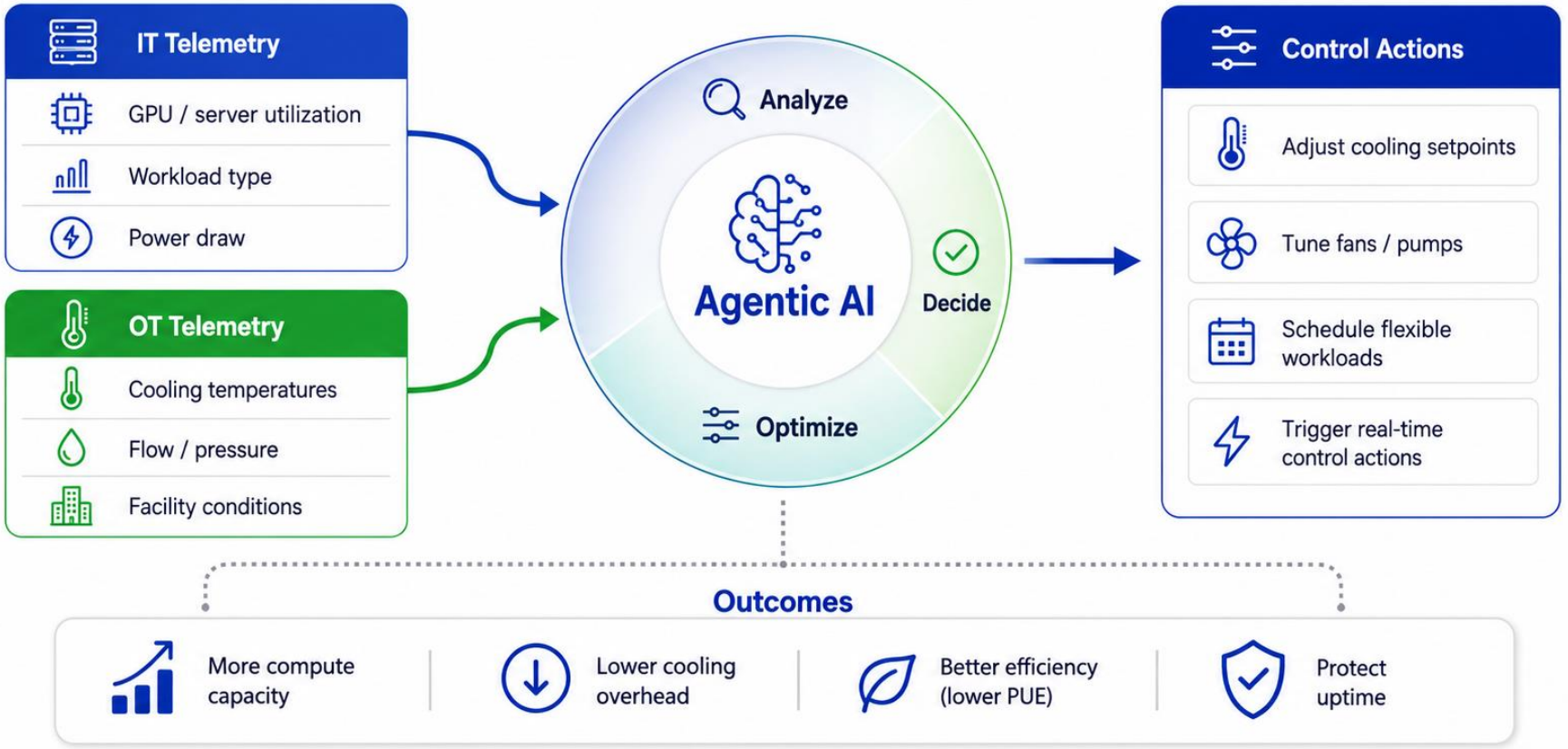
Grid interconnects and permitting cap the **total**. Adding megawatts is a multi-year fight — moving the divider is not.
Every point shifted out of cooling is a point that does useful work — on the same grid connection.

WHAT "MORE TOKENS, SAME MEGAWATTS" MEANS

Cooling megawatts saved → **Compute megawatts gained** → **More tokens per day, same power bill- higher Tokens/watt and ROI**

You can't widen the budget — you can only decide how much of it does useful work

IT + OT Convergence: One Agent, One Decision

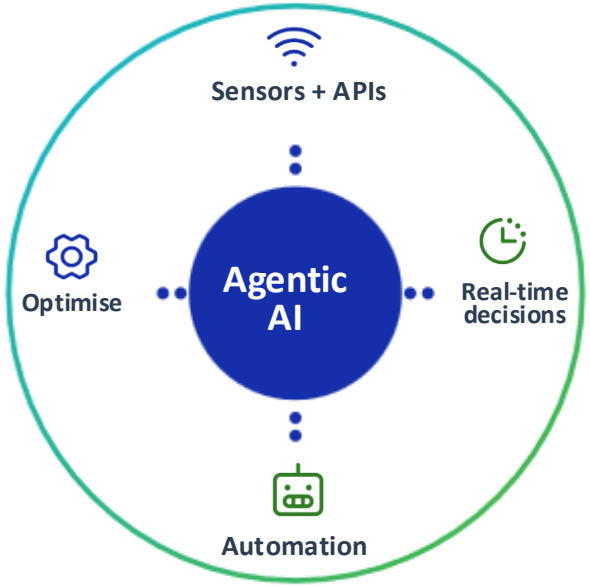


The same telemetry split our own environment observes — IT signals from the chip, OT signals from the plant, one agent deciding.

Redirecting Cooling Savings to Compute Delivers High ROI and Token/watt

Power | IT compute | Cooling | AI control | AI output

BEFORE	
Total power	500 MW
IT compute	325 MW
Cooling load	175 MW
PUE	1.54
Tokens per day	363T



AFTER		
Total power	500 MW	no change
IT compute	425 MW	↑ +31%
Cooling load	75 MW	↓ −57%
PUE	1.18	↓ −23%
Tokens per day	476T	↑ +31%

0% change in power bill

+31% compute capacity

−57% cooling overhead

+31% AI output per day

Illustrative model at 500 MW total facility power; percentages are change from the before case.

So We Built the Thing and Tested It

A chip-to-chiller data center simulation, with AI agents built on LLMS making real cooling decisions



One Coupled Simulator: Chip to Chiller

Every watt the silicon burns is followed through four coupled physical models until it leaves the building as warm air.

JUNCTION

AMBIENT AIR

STAGE 01
Chip

NVIDIA B200 — 32×32 RC thermal grid with DVFS governor.

Calibrated to real 8×B200 telemetry

STAGE 02
Cold plate

Copper heat-spreading model with thermal-mass lag and a per-channel convective energy balance.

STAGE 03
CDU

Pump and heat-exchanger loop scaled from a real CDU: NTU-effectiveness transfer, pump affinity-law flow.

STAGE 04
Air-cooled chiller

JCI York YVAM magnetic-bearing centrifugal: real capacity/EIR curves against CHWS and ambient dry-bulb, fan-driven heat rejection.

COOLANT RETURNS — THE LOOP IS CLOSED, EVERY TIMESTEP

3,360 GPUs simulated

420 nodes × 8 B200s

2 MW IT load

Three Experiment Configurations: Actions × Agents

SCENARIO	ACTION(S)	AGENT(S)
1	Chiller CHWS setpoint only	Single agent
2	Chiller CHWS setpoint + CDU pump speed	Single agent
3	Chiller CHWS setpoint + CDU pump speed	Two agents

All scenarios: CHWS range 15°C–30°C, decisions every 600 seconds (10 minutes) -- only the action space and agent count vary.

Local agent compute : Scenario 1 ≈ 10 W · Scenario 2 ≈ 67.5 W · Scenario 3 ≈ 67.5 W — negligible next to the ~2 MW IT load.

The Agent's Only Goal: Tokens per Joule

Reward = Tokens generated / Energy consumed

The agent is scored on the exact currency the business cares about -- “more tokens, same megawatts” is its literal objective function, not a slogan layered on afterward.

Safety trip: episode ends immediately at a real 92°C junction-temperature limit -- a physical constraint, not a penalty term.

Eight Models, One Closed Loop

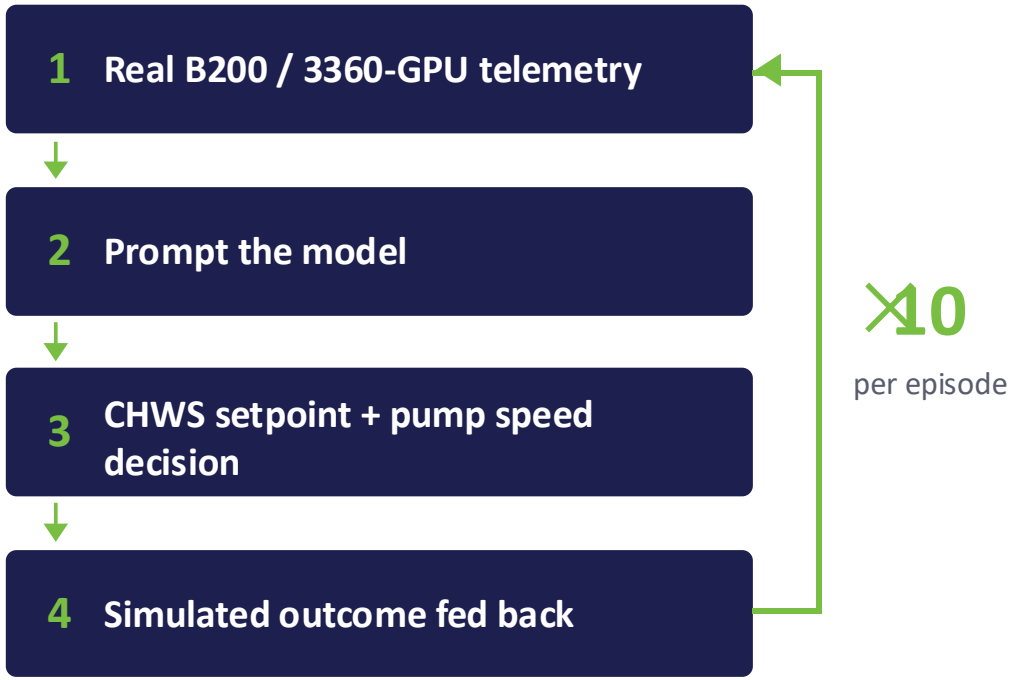
Zero-shot LLM agents: no training or fine-tuning. Every decision returns plain-language reasoning.

THE MODELS

Local via Ollama	Parameters
llama3.1	8B
qwen3	8B
qwen2.5-instruct	7B
phi3.5	3.8B
qwen2.5	3B
gemma2	2B

Cloud API	
GPT-5.4-Mini	API
Claude-Haiku-4-5	API

THE CLOSED LOOP



Held constant across all eight: Environment | Action space | Reward

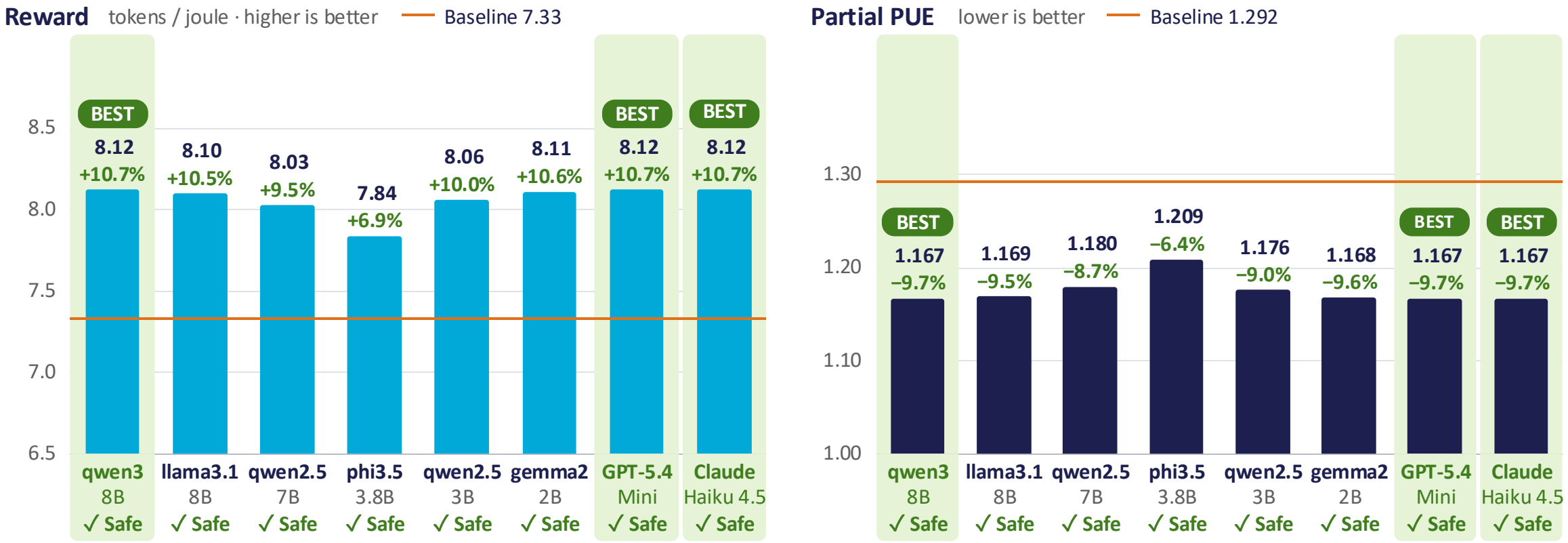
10 decisions × 600 s = 6,000 s per episode

Scenario 1 · Single Action (Chiller Setpoint) · Single Agent

Scenario 1 Results: Chiller Setpoint, One Agent

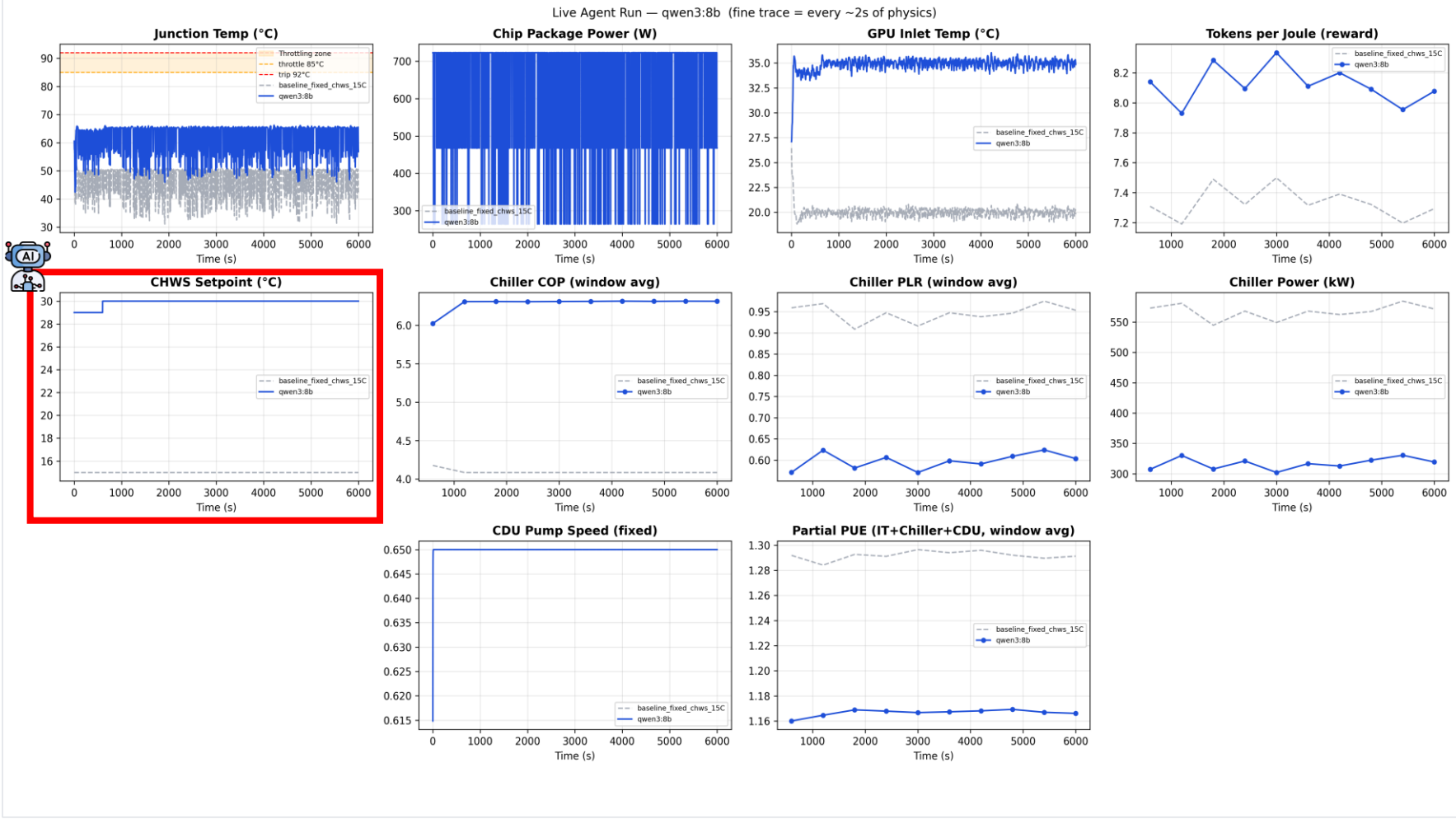
All eight agents beat the fixed-setpoint baseline, and none crosses 85 °C.

BEST Best agent Light bars = exceeded 85 °C



Baseline: CHWS fixed at 15 °C. Air-cooled chiller, B200 / 3360-GPU simulator. 6 agents local via Ollama, 2 via cloud API.

Scenario 1 Live Run: qwen3:8b



Stays below 85 °C

+10.7%

reward vs. baseline

–9.7%

Partial PUE

–44.1%

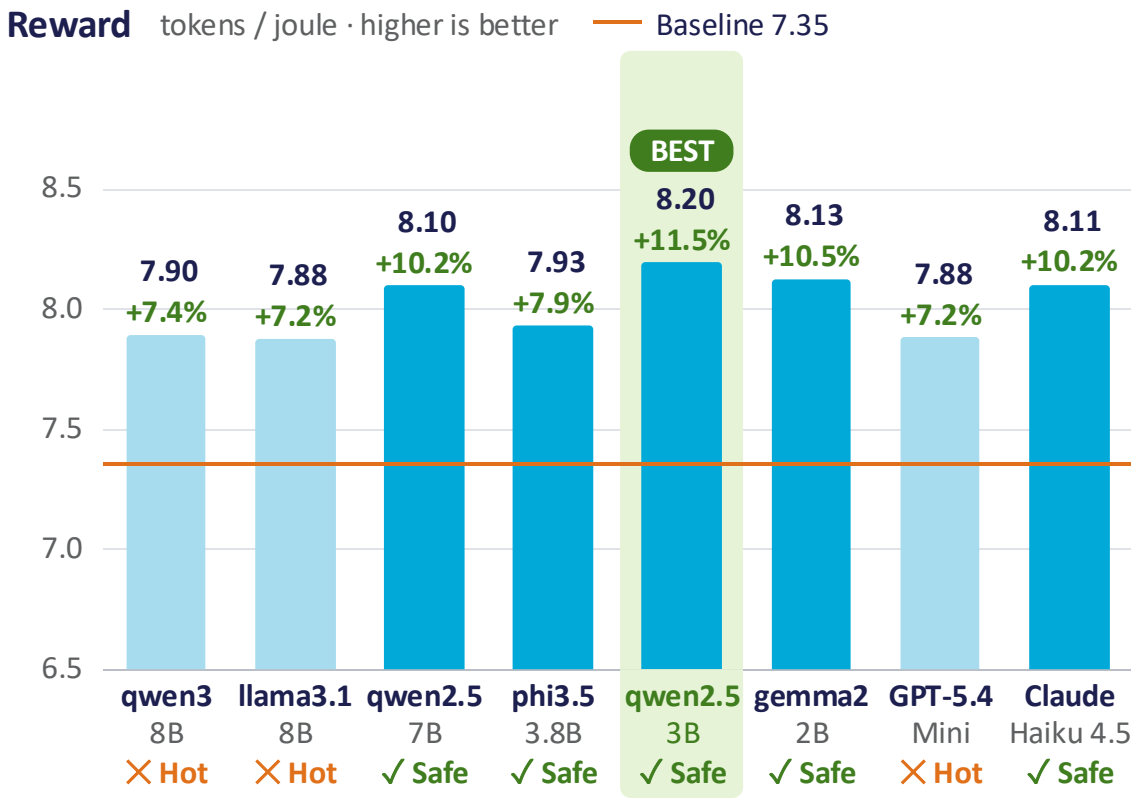
chiller power

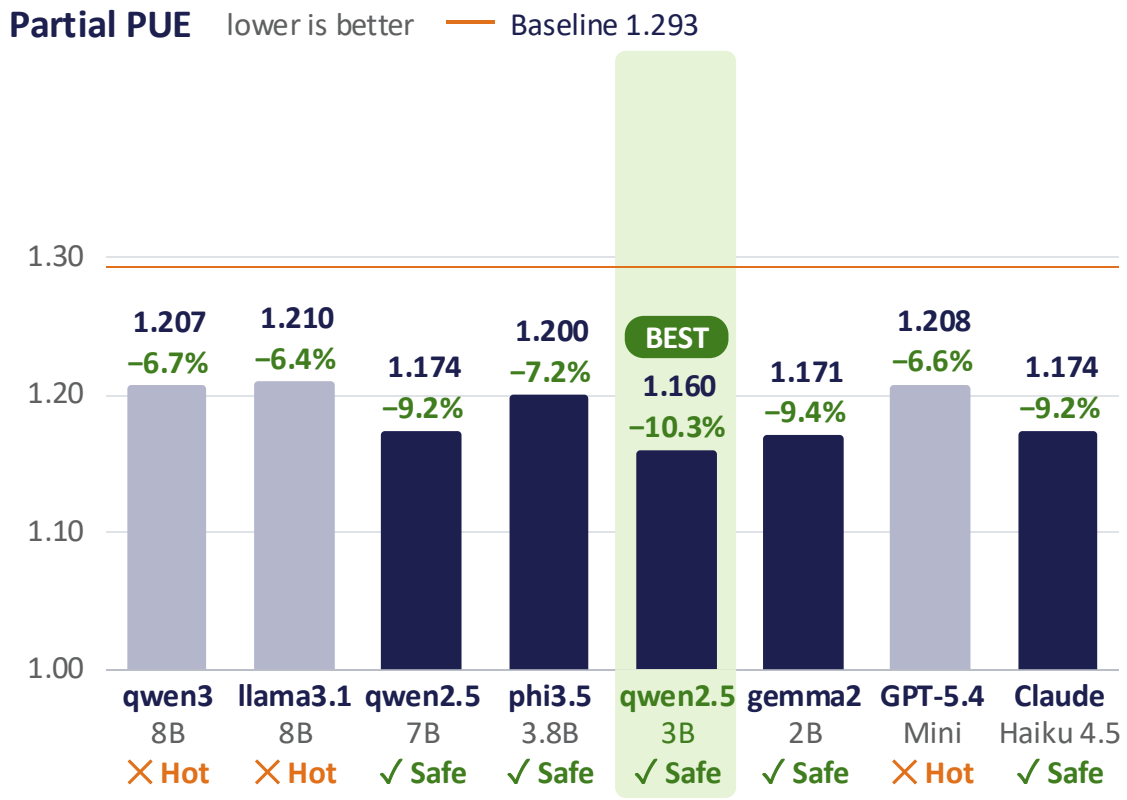
Scenario 2 · Two Actions (Chiller Setpoint + Pump Speed) · Single Agent

Scenario 2 Results: Setpoint + Pump, One Agent

Adding pump control lifts the best reward to +11.5%, but three agents now overheat.

BEST Best agent Light bars = exceeded 85 °C





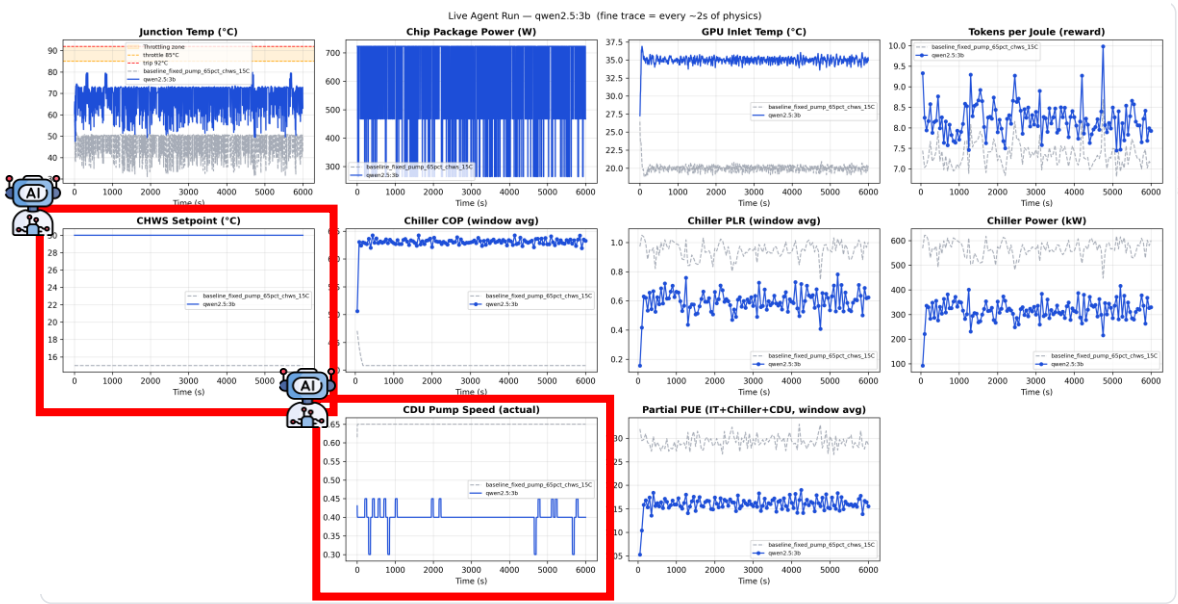
Baseline: CHWS fixed at 15 °C, pump at 65%. Air-cooled chiller, B200 / 3360-GPU simulator. 6 agents local via Ollama, 2 via cloud API.

Scenario 2 Live Runs: Safe vs. Overheating

Full 10-signal dashboards over the 6,000 s episode, under identical conditions.

qwen2.5:3b

Stays below 85 °C



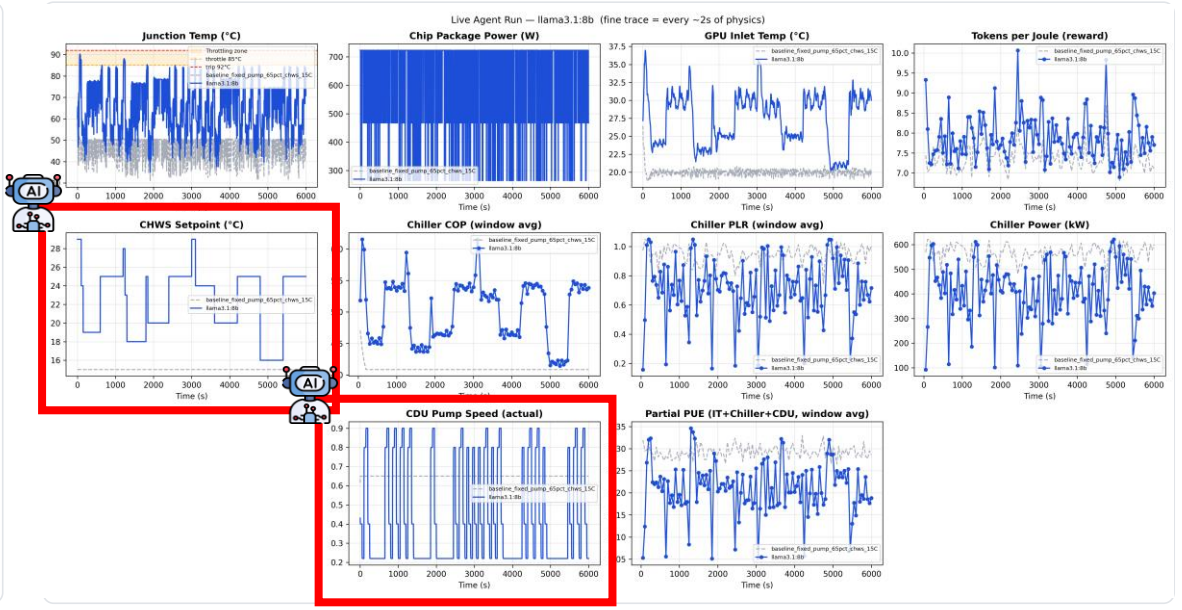
+11.5% reward

−44.3% chiller power

−38% pump speed

llama3.1:8b

Exceeds 85 °C



+7.2% reward, lowest of 8

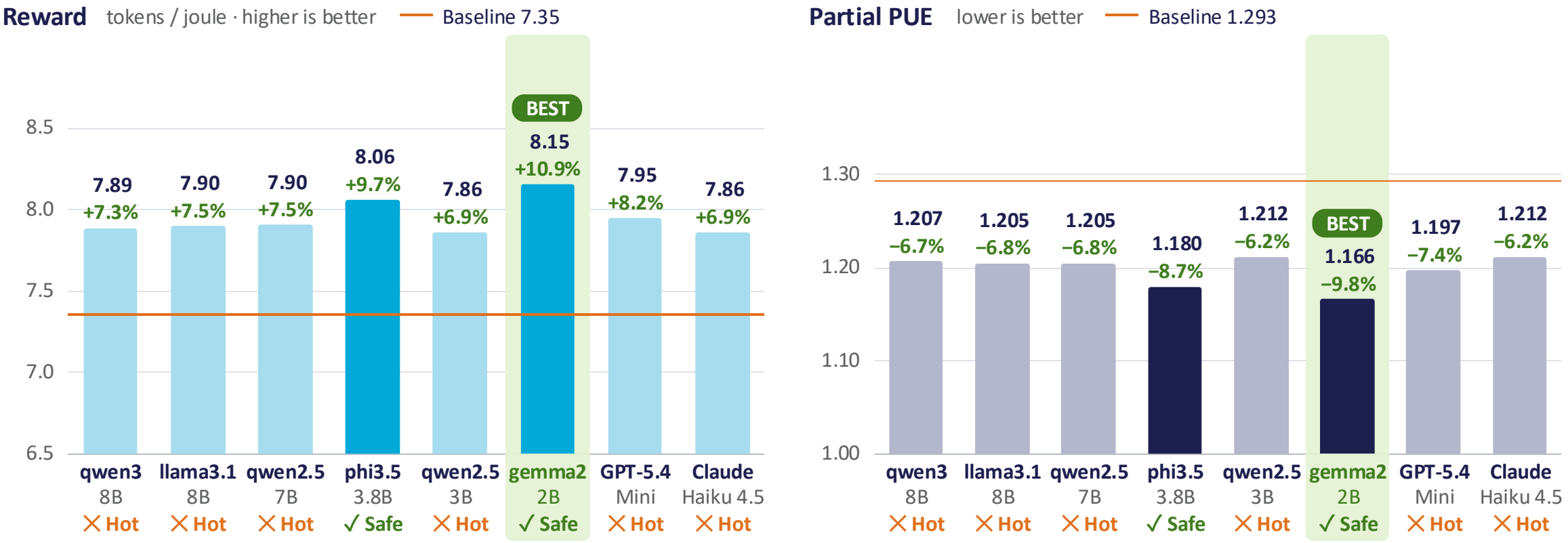
−4.3 pts vs. qwen2.5:3b

Scenario 3 · Two Actions (Chiller Setpoint + Pump Speed) · Two Agents

Scenario 3 Results: Setpoint + Pump, Two Agents

Splitting control keeps the efficiency gains, but six of eight agents overheat.

BEST Best agent Light bars = exceeded 85 °C



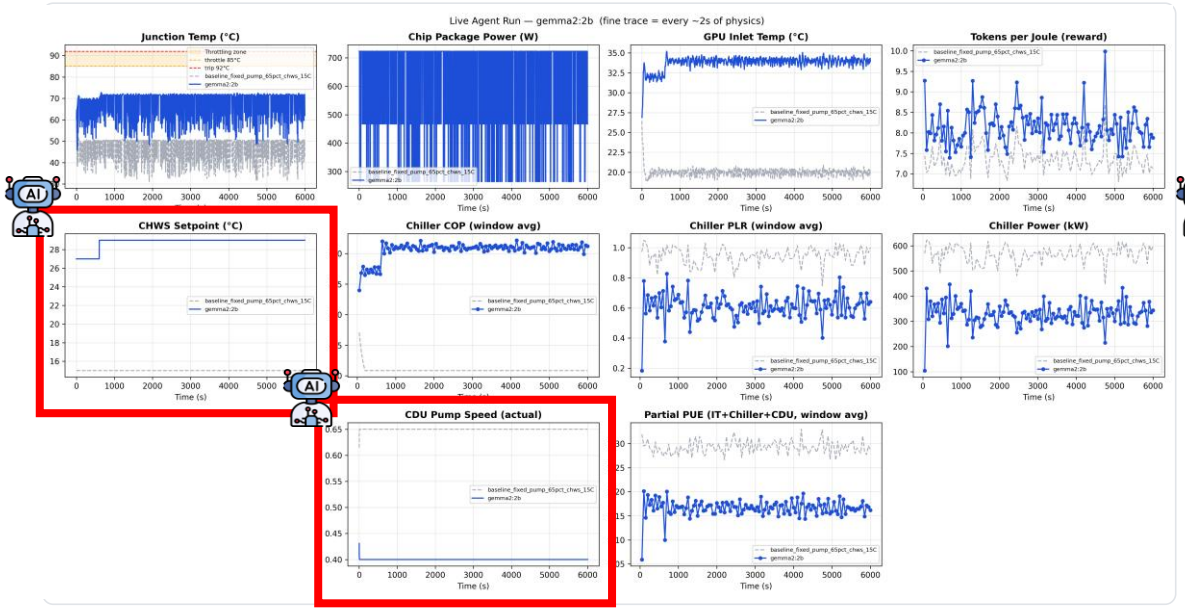
Baseline: CHWS fixed at 15 °C, pump at 65%. Air-cooled chiller, B200 / 3360-GPU simulator. 6 agents local via Ollama, 2 via cloud API.

Scenario 3 Live Runs: Safe vs. Overheating

Full 10-signal dashboards over the 6,000 s episode, under identical conditions.

gemma2:2b

Stays below 85 °C



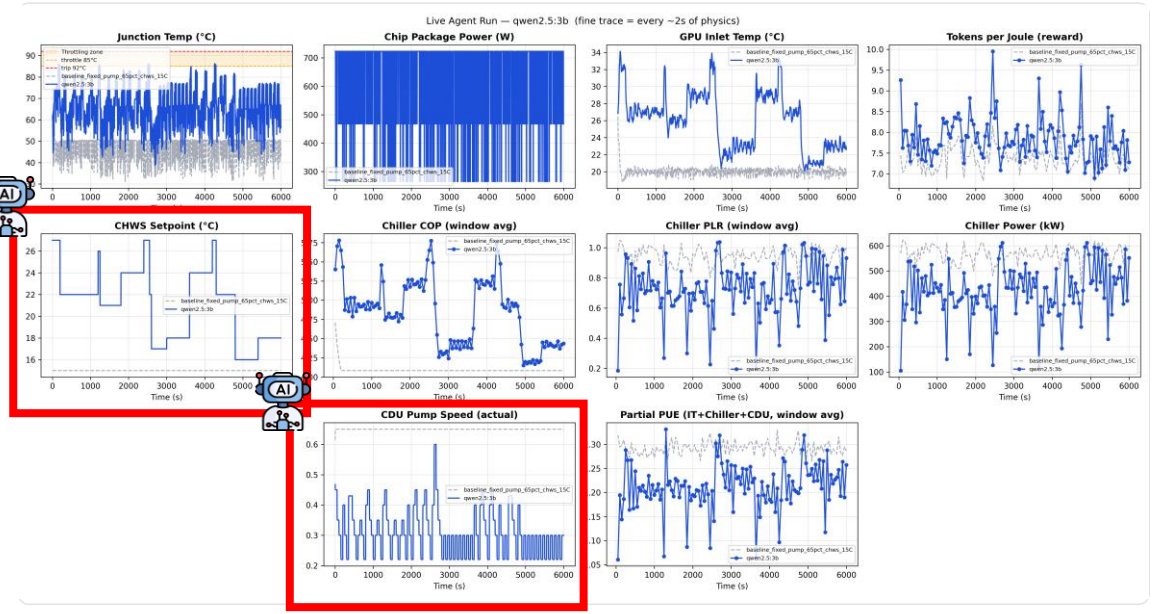
+10.9% reward

−42.0% chiller power

−38.5% pump speed

qwen2.5:3b

Exceeds 85 °C



+6.9% reward, tied lowest

−4.0 pts vs. gemma2:2b

SUMMARY · THREE SCENARIOS · EIGHT AGENTS

Every agent saved energy. Most could not stay cool.

Reward and Partial PUE improve in all three scenarios, for all eight agents. Thermal safety does not follow.

WITH TWO ACTIONS AND TWO AGENTS

2 of 8 models

stay under the 85°C junction-temperature throttle down from 8 of 8 with a single action.

The six that overheat still cut Partial PUE by 6–7%. Efficiency metrics alone would have passed them.

SCENARIO BY SCENARIO

Single action · single agent

Chiller setpoint

8 of 8 SAFE

Reward +6.9 → +10.7% **pPUE** −6.4 → −9.7%

Two actions · single agent

Setpoint + pump speed

5 of 8 SAFE

Reward +7.2 → +11.5% **pPUE** −6.4 → −10.3%

Two actions · two agents

Setpoint + pump speed, split control

2 of 8 SAFE

Reward +6.9 → +10.9% **pPUE** −6.2 → −9.8%

BEST AGENT IN EACH SCENARIO IS ALSO SAFE

qwen3:8b · qwen2.5:3b · gemma2:2b — all open-weight, all local. Safety is not the price of performance.

Scaling to One Gigawatt AI Factory

Same lever. Not yet the same scale.

TODAY

2 MW

3360 GPUs · proven

→

GIGAWATT ERA

GIGA-SCALE

same lever, unproven at this scale

8	3	+11.5%	-10.3%
LLM AGENTS TESTED	CONTROL SCENARIOS	PEAK REWARD GAIN	PEAK PUE CUT

Best Performer, Safest Model

In every scenario tested, the top-scoring model on reward also stayed below the 85°C throttle threshold — performance and safety point the same direction here.

Efficiency Gains Become Compute Gains

Every megawatt saved by cutting PUE isn't wasted — under a fixed power budget, it can be redirected straight into more IT compute capacity.

AHEAD OF US Gigawatt-scale validation · Real BAS integration · Safety-aware reward design · Multi-episode validation

More Tokens. Same Megawatts. Today.

8 AGENTS | 3 SCENARIOS | +11.5% REWARD | -10.3% PUE

1. Partner on scaling the demo to multi-rack, real-BAS-integrated deployments
2. Collaborate through OCP working groups on thermal / IT telemetry convergence and safety-aware reward design
3. Compare notes — we're sharing the simulator's design and our thermal-safety findings, not just the headline numbers

Call to Action

- **Problem to solve**
  - Cooling control and IT telemetry are still optimized separately, so efficiency gains stop at the facility boundary instead of becoming compute.
- **How to get involved**
  - Partner on scaling the demo to multi-rack, real-BAS-integrated deployments
  - Collaborate through OCP working groups on thermal / IT telemetry convergence and safety-aware reward design
  - Compare notes — we're sharing the simulator's design and our thermal-safety findings, not just the headline numbers

Thank You!

